

4(a)	<i>allow points from correctly labelled diagram</i> (monomers of) beta-glucose / β-glucose ; R α-glucose / two β-glucoses <i>plus any two from:</i> <i>max 1 if no monomer stated</i> (joined by) 1, 4 glycosidic bonds ; A 1,4 glucosidic bonds R 1,4 <u>and</u> 1,6 bonds each (β-) glucose rotated by 180° compared to adjacent (β-)glucose ; AW linear / straight, chain / molecule; unbranched ;	3
4(b)(i)	<i>any one from:</i> increases rate of translation of, enzyme / nitrate reductase ; increases, enzyme / nitrate reductase, synthesis / concentration ; A more nitrate reductase produced <i>idea of</i> more metabolically efficient ; e.g. enzyme only produced when needed AVP ; e.g. more nitrate is reduced to nitrite / more nitrite is formed leads to increased supply of nitrite for amino acids / amino acid synthesis	1

Question	Answer	Marks
4(b)(ii)	<i>any two from:</i> improves accuracy of results / results not by eye / results not subjective ; A detects smallest change in, (intensity of) colour / concentration / quantity provides quantitative results / obtain numerical values ; A described e.g. absorbance values / can measure absorbance/ gives percentage transmission / absorbance readings <i>idea that</i> different, intensities / shades of, colour / magenta / red-purple, relate to different, quantities / concentrations, of nitrites ; A relate to (rates of) nitrate uptake can produce calibration curve (for quantitative results) ; AVP ; e.g. can detect, (very) low concentrations / faintly coloured samples	2

Question	Answer	Marks
4(c)(i)	<i>accept hydrogen ions / H⁺, for protons</i> <i>max 3 if no ref. to amino acids</i> <i>any four from:</i> 1 protons, moved by active transport / pumped, out of (companion) cell ; A using, energy / ATP <i>for active transport</i> 2 protons, moved into / enter / AW, cell wall / apoplast ; R if moved out to phloem sieve tube 3 proton gradient builds up in, apoplast / cell wall ; A higher concentration A high concentration if mp4 gained 4 protons move back (into companion cell), by facilitated diffusion / AW ; e.g. down, electrochemical / concentration, gradient from high(er) to low(er) concentration A diffuse <i>if movement is described through a transport protein in mp5</i> 5 protons cotransport amino acids or amino acids move with protons through, a cotransporter / a cotransport protein / (amino acid) transporter ; 6 amino acids, transported / AW, against their concentration gradient ;	4
4(c)(ii)	plasmodesmata present ; A movement is down the, concentration / diffusion, gradient A because there is a high(er) concentration in the companion cell ora for phloem sieve tube (sap)	1

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4(d)	0.2 mmol dm⁻³ constant rate v 5.0 mmol dm⁻³, rate not constant / described ; 0.2 mmol dm⁻³ lower rate than 5.0 mmol dm⁻³ ; ora 5.0 mmol dm⁻³ higher rate A faster / slower, rate for higher / lower, rate A (overall) steeper rate / less steep rate calculated rates to support ;	2										
4(e)	any four from: 1 (experiments suggest mainly) active transport / active uptake ; A uptake, is an active process / needs energy / needs ATP <i>if mp8 given, allow 'active transport and facilitated diffusion'</i> 2 comparison of experiments with control, any one ; <table><tr><th>expt</th><th>discussion point</th></tr><tr><td>1</td><td>oxygen required for (complete) uptake A aeration required or almost no / AW, uptake, with no oxygen / in anaerobic conditions</td></tr><tr><td>2</td><td>almost no / AW, uptake, in, low temperature / at 3°C or uptake, higher / AW, at 30°C / in control</td></tr><tr><td>3</td><td>proteins / enzymes, are, required for / involved in, uptake or (much) lower / AW, uptake when, proteins not synthesised / protein synthesis inhibitor present</td></tr><tr><td>4</td><td>slightly lower / very similar, uptake / AW, without any bacteria or slightly higher / very similar, uptake / AW, with bacteria (control)</td></tr></table> 3 one comparative result / manipulated data, with units, to support mp2 ;	expt	discussion point	1	oxygen required for (complete) uptake A aeration required or almost no / AW, uptake, with no oxygen / in anaerobic conditions	2	almost no / AW, uptake, in, low temperature / at 3°C or uptake, higher / AW, at 30°C / in control	3	proteins / enzymes, are, required for / involved in, uptake or (much) lower / AW, uptake when, proteins not synthesised / protein synthesis inhibitor present	4	slightly lower / very similar, uptake / AW, without any bacteria or slightly higher / very similar, uptake / AW, with bacteria (control)	4
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4(e)	experiment	experiment v control	or	lower by	only needs to state control value once somewhere in response		
	1	0.4 v 10.9 μmol g ⁻¹		10.5 μmol g ⁻¹			
	2	0.6 v 10.9 μmol g ⁻¹		10.3 μmol g ⁻¹			
	3	1.4 v 10.9 μmol g ⁻¹		9.5 μmol g ⁻¹			
	4	10.0 v 10.9 μmol g ⁻¹		0.9 μmol g ⁻¹			
	discussion points						
	4 aerobic respiration needs oxygen ;						
	5 lower temperature means lower activity of enzymes involved in respiration ;						
	6 ref. to ATP (for active transport) ; e.g. (aerobic) respiration produces ATP less respiration means less ATP produced no ATP for, conformational change / AW (of carrier protein)						
	7 some (facilitated) diffusion because some uptake without oxygen ;						
	8 protein synthesis needed for transport proteins ; A membrane / transport / carrier / channel, proteins not being synthesised A no protein synthesis so less ATP synthase						
	9 effect of lower temperature on transport ; ora for control e.g. reduces rate of (facilitated) diffusion affects, ATPase / carrier protein, for active transport transport proteins less efficient at low temperature						
	10 bacteria will have, little / no, effect on uptake or (normally) bacteria (associated with roots), increase / help, nitrate uptake ;						
	11 AVP ; e.g. idea that nitrate uptake involves metabolic processes						